D75-D76 | Information cannot finance a choice

SYNTHETIC private classroom lab. All payoffs, probability, scores, signals and review roles are invented. No real research purchase, financing, contract or consent.

Read the units	All monetary endpoints are decision-date present values in lakh INR over the same 12-month horizon. Already net of original case investment and operating costs. Do not subtract upfront requirements again from these endpoints.
Keep the two tests separate	Decisions first ranks risk-neutral monetary EV without funding constraints, then screens upfront needs against current cash. Passing that day-0 screen does not establish full cash-path safety, quality or authority.
Perfect information	State-contingent best choices are recomputed for the same feasible action set. At cash 6, build requiring 10 is unavailable. Partner dominates defer in either state, so market-state information alone has zero gross value in that restricted set.
Invented imperfect signal	Signal adds NEW hypothetical conditional accuracy P(+|favorable)=0.8 and P(+|unfavorable)=0.2. These are not measured pilot results. Joint probabilities, posteriors and state-weighted payoffs are shown. A positive result is not certainty.
Research changes cash	Research cost is new, outside the already-net original endpoints. Pay it once in the modeled payoff and deduct it once from available day-0 cash before screening actions. This simplified timing assumption is invented, not verified financing.
Changed-context retest	Change p to 0.50 and restore it; then change cash to 2 with research cost 2. Compare the same action sets before/after information. Zero-probability signals have undefined posteriors but zero branch contribution. Restore inputs.
No model is approval	Matrix points stay separate from lakh and hard gates. Day 0 evidence role checks inputs; day 30 funding role checks cash path; day 60 terms role checks cost/delay/exit rights; day 90 sponsor role reviews or defers. All roles fictional.

Inputs | Label assumptions before calculating

Original D75-D76 payoffs and liquidity screen retained. Signal accuracy is an explicitly invented extension; no current empirical evidence or extra authority is claimed.

Favorable-state probability	0.65
Build favorable net payoff	12.00
Build unfavorable net payoff	-8.00
Partner favorable net payoff	6.00
Partner unfavorable net payoff	1.00
Build upfront need	10.00
Partner upfront need	2.00
Available uncommitted cash	6.00
New research cost, paid before action	2.00
P(positive | favorable)	0.80
P(positive | unfavorable)	0.20
Cost weight	0.40
Time weight	0.30
Control weight	0.30

Inputs valid? (1=yes)	1.00

Payoffs may legitimately be negative. Cash, upfront needs and research cost must be nonnegative; probabilities 0..1; weights sum to 1. Yellow entries are editable, but validity checks structure only. No financing, permission, evidence accuracy or full cash-path validation follows.

Decisions | EV, feasibility and permission differ

Risk-neutral monetary teaching model. Defer is always zero with zero upfront need in this case only. Full cash path, terms and real owner permission remain unverified.

Choice	Favorable	Unfavorable	EV at current p	Upfront need	Day-0 feasible
Build	12.00	-8.00	5.00	10.00	0.00
Partner	6.00	1.00	4.25	2.00	1.00
Defer	0.00	0.00	0.00	0.00	1.00

Best no information, unrestricted	5.00
Perfect information, unrestricted	8.15
Gross EVPI, unrestricted	3.15
Perfect-info payoff minus research cost	6.15
Net gain versus no research	1.15

Best no information, cash-screened	4.25
Perfect information, cash-screened	4.25
Gross EVPI, same cash-screened set	0.00

Build funding shortfall	4.00
Build / partner crossover p	0.60

Baseline: build EV 5, partner 4.25; unconstrained EVPI 3.15. Build has an 8 unfavorable loss and 4 upfront shortfall. Perfect-information cost bridge is hypothetical and unconstrained; it is NOT the imperfect-signal value or an approved budget. Crossover may lie outside [0,1], or be undefined for parallel EV lines.

Sensitivity | Change probability, not the units

Every row recomputes decisions and perfect-information value with the same current payoffs, cash and action set. Equal EVs are ties, not a hidden preference.

p favorable	Build EV	Partner EV	Unrestricted best	Gross EVPI	Cash-screened best	Screened EVPI
0.00	-8.00	1.00	1.00	0.00	1.00	0.00
0.05	-7.00	1.25	1.25	0.30	1.25	0.00
0.10	-6.00	1.50	1.50	0.60	1.50	0.00
0.15	-5.00	1.75	1.75	0.90	1.75	0.00
0.20	-4.00	2.00	2.00	1.20	2.00	0.00
0.25	-3.00	2.25	2.25	1.50	2.25	0.00
0.30	-2.00	2.50	2.50	1.80	2.50	0.00
0.35	-1.00	2.75	2.75	2.10	2.75	0.00
0.40	0.00	3.00	3.00	2.40	3.00	0.00
0.45	1.00	3.25	3.25	2.70	3.25	0.00
0.50	2.00	3.50	3.50	3.00	3.50	0.00
0.55	3.00	3.75	3.75	3.30	3.75	0.00
0.60	4.00	4.00	4.00	3.60	4.00	0.00
0.65	5.00	4.25	5.00	3.15	4.25	0.00
0.70	6.00	4.50	6.00	2.70	4.50	0.00
0.75	7.00	4.75	7.00	2.25	4.75	0.00
0.80	8.00	5.00	8.00	1.80	5.00	0.00
0.85	9.00	5.25	9.00	1.35	5.25	0.00
0.90	10.00	5.50	10.00	0.90	5.50	0.00
0.95	11.00	5.75	11.00	0.45	5.75	0.00
1.00	12.00	6.00	12.00	0.00	6.00	0.00

Baseline crossover p=0.60. At p=0.50: build 2, partner 3.5. At 0.65: build 5, partner 4.25. The cash screen still excludes build at cash 6. Scores, funding or a different risk objective can change the actual choice; no automatic release.

Signal | An invented test is not perfect information

NEW synthetic extension: conditional signal probabilities, immediate cost before action, no delay/options change. No pilot data, observed accuracy or actual research approval.

Signal	Joint favorable	Joint unfavorable	Signal probability	Posterior favorable	Build joint value	Partner joint value	Best joint value
Positive	52.00%	7.00%	59.00%	88.14%	5.68	3.19	5.68
Negative	13.00%	28.00%	41.00%	31.71%	-0.68	1.06	1.06

Joint mass sums to one	1.00
EV with sample information, unrestricted	6.74
Gross EVSI, unrestricted	1.74
After research cost, unrestricted	4.74
Net change vs no research	-0.26

Cash after research cost	4.00
Research affordable now? (1=yes)	1.00
Build feasible after cost? (1=yes)	0.00
Partner feasible after cost? (1=yes)	1.00

Research branch EV, funded actions	4.25
Funded research payoff after cost	2.25
Net change vs cash-screened baseline	-2.00

Baseline P(+)=0.59; posterior favorable after + is 0.52/0.59, not 1 or prior 0.65. Gross EVSI 1.74 <= EVPI 3.15. Cost 2 makes unrestricted net change -0.26. Under cash 6, partner remains the only nondefer funded choice after cost; net change -2. Zero-probability signal posterior is undefined, but its joint contribution is zero. Delay and full cash paths are absent, so this is not a research recommendation.

Matrix | Points are not lakh

Synthetic subjective scores, higher better, 1..5. Weights from Inputs sum to 1. Scores do not override funding, safety, permission or contract gates.

Alternative	Cost score	Time score	Control score	Weighted points	Scores valid
Build	2.00	2.00	5.00	2.90	1.00
Partner	4.00	4.00	2.00	3.40	1.00
Defer	5.00	1.00	1.00	2.60	1.00

Proposed scale anchors: cost score 1=least attractive to 5=most attractive; time 1=slowest to 5=fastest; control 1=least to 5=most. These invented relative judgments are not measured money or universal interval scales. Check criterion overlap and test weights. Baseline points build 2.9, partner 3.4, defer 2.6 are not added to monetary EV.

Failure log | Observed authored tests

Not learner attempts, independent review, Excel/Sheets parity, measured transfer or real business authorization.

Test / context	Observed authored outcome
Baseline exact calculations	PASS: EV, cash screen, EVPI, every probability row, conditional signal arithmetic and matrix matched independent exact-rational calculations.
Probability 0.50 transfer	PASS: p 0.50 gives build 2 versus partner 3.5; all branches and sensitivity checked; restored.
Funding cash 2 transfer	PASS: cash 2 can fund partner before research; cost 2 leaves only defer, funded research payoff -2 and change -6.25. Restored.
Uninformative signal	PASS: uninformative signal keeps posterior at prior and gross EVSI zero; cost still reduces payoff; restored.
Perfect signal	PASS: perfect conditional signal attains unconstrained EVPI 3.15, not more; funded action set remains separate; restored.
Impossible signal branch	PASS: impossible positive signal has undefined posterior and zero joint payoff, not fake certainty. Remaining branch tested; restored.
Invalid probability / inputs	PASS: negative/out-of-range/text/blank inputs suppressed numeric results. Cost 7 with cash 6 made research unavailable, not affordable; restored.
Bad weights / scores	PASS: weights not summing to 1 and score outside 1..5 flagged; no fake valid point total; restored.
Parallel payoff slopes	PASS: parallel build/partner payoff slopes leave crossover undefined rather than divide by zero; restored.
Broken reference	PASS: broken joint-probability reference propagated to sample-information results, not a false numeric pass; restored.
Restored baseline	PASS: all baseline formula values restored. Authored native execution only, not independent learner mastery or spreadsheet-app parity.

Keep your own attempted reasoning, error/cause, correction and delayed changed-context retest. Source: original synthetic D75-D76 at https://top1os.dev/ ; new signal/timing assumptions explicitly authored. Existing MIT decision-tree/value-of-information readings remain framing, not empirical probability or financial permission.